

Disproof of Conjecture 000000000199 (quadratic-denominator psi representation)

TLMC falsification package

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Conjecture 000000000199. *Let $S = \sum_{n \geq 0} \frac{1}{n^2 + n + 2}$, whose denominator is irreducible over $\mathbb{Q}$. The sum is an explicit $\mathbb{Q}$ -linear combination of values of the digamma function at cube roots of unity; the irrationality of S is equivalent to that of $\psi(\omega)$ -type values ($\omega^3 = 1$), the equivalence following from an explicit partial-fraction decomposition.*

Verdict: FALSE. The partial-fraction decomposition invoked by the conjecture produces digamma values at the roots of $z^2 + z + 2$, namely $(-1 \pm i\sqrt{7})/2$ — points of modulus $\sqrt{2} \neq 1$, lying in the discriminant -7 quadratic field — and *not* at cube roots of unity, which belong to the discriminant -3 world of $z^2 + z + 1$.

1 The correct decomposition

Write $n^2 + n + 2 = (n + a)(n + b)$ with

$$a = \frac{1 + i\sqrt{7}}{2}, \quad b = \frac{1 - i\sqrt{7}}{2}, \quad a + b = 1, \quad ab = 2, \quad a - b = i\sqrt{7}.$$

The digamma identity $\psi(v) - \psi(u) = \sum_{n \geq 0} \left(\frac{1}{n+u} - \frac{1}{n+v} \right)$ gives

$$S = \sum_{n \geq 0} \frac{1}{(n+a)(n+b)} = \frac{1}{b-a} \sum_{n \geq 0} \left(\frac{1}{n+a} - \frac{1}{n+b} \right) = \frac{\psi(b) - \psi(a)}{b-a}. \quad (1)$$

By Schwarz reflection ($\psi(\bar{z}) = \overline{\psi(z)}$),

$$S = \frac{2}{\sqrt{7}} \operatorname{Im} \psi\left(\frac{1 + i\sqrt{7}}{2}\right) = 1.1868273377200538821608430611187162259569 \dots \quad (2)$$

Equivalently, since $\psi(z+1) = \psi(z) + 1/z$, the arguments are the roots $r_{\pm} = (-1 \pm i\sqrt{7})/2$ of $z^2 + z + 2$ shifted by 1.

Numerically, (1) and a direct summation of $2 \cdot 10^5$ terms with a digamma tail agree to 6.4×10^{-14} in double precision, and to $\sim 10^{-39}$ in 40-digit arithmetic (see `reproduce.py`); so (2) is the true representation of S .

2 The attack: the arguments are not cube roots of unity

Claim 1. *The roots of $z^2 + z + 2$ are $r_{\pm} = (-1 \pm i\sqrt{7})/2$ with $|r_{\pm}|^2 = r_+ \bar{r}_+ = \frac{1+7}{4} = 2 \neq 1$, and $r_+^3 = \frac{5-i\sqrt{7}}{2} \neq 1$. The discriminant of $z^2 + z + 2$ is $b^2 - 4ac = 1 - 8 = -7$.*

Claim 2. *Every cube root of unity is unimodular and is a root of $z^2 + z + 1$ or $z - 1$, polynomials over the discriminant -3 field: the primitive ones satisfy $\omega^2 + \omega + 1 = 0$ with discriminant $1 - 4 = -3$, $|\omega| = 1$, $\omega^3 = 1$.*

Since $-7 \neq -3$, the values $\psi\left(\frac{1 \pm i\sqrt{7}}{2}\right)$ produced by the exact decomposition (1) are digamma values at non-unimodular, non-root-of-unity arguments. The conjecture asserts that (1)-type decomposition yields ψ at *cube roots of unity*; for this denominator it does not, and no mechanism replaces $\psi\left(\frac{1 \pm i\sqrt{7}}{2}\right)$ by $\mathbb{Q}$ -combinations of $\psi(1), \psi(\omega), \psi(\omega^2)$. The claimed equivalence of the irrationality of S with that of $\psi(\omega)$ -type values is therefore unsupported and false as stated.

Numerical corroboration. $\psi(1) = -\gamma = -0.57721566490\dots$ and $\psi(\omega) = 0.28507344127\dots + 2.42326662850\dots i$ with $\text{Im } \psi(\omega) \neq 0$. Hence any *real* $\mathbb{Q}$ -linear combination $q_1\psi(1) + q_2\psi(\omega) + q_3\psi(\omega^2)$ must have $q_2 = q_3$, i.e. lies in $\text{span}_{\mathbb{Q}}\{\psi(1), 2\text{Re } \psi(\omega)\}$. An exhaustive search over denominators ≤ 500 finds no such combination matching $S = 1.18682733772005\dots$ (best residuals $> 10^{-12}$), consistent with the verdict.

3 Boundary: why the mechanism works for discriminant -3 only

For the twin denominator $n^2 + n + 1$ the same decomposition gives

$$\sum_{n \geq 0} \frac{1}{n^2 + n + 1} = \frac{\psi(\bar{\alpha}) - \psi(\alpha)}{\bar{\alpha} - \alpha} = \frac{2}{\sqrt{3}} \text{Im} \psi\left(\frac{1 + i\sqrt{3}}{2}\right) = 1.7981472805626901809\dots,$$

and here $\alpha = \frac{1+i\sqrt{3}}{2}$ is a sixth root of unity ($\alpha^6 = 1$, $|\alpha| = 1$, disc -3), so the cube-root-of-unity representation genuinely applies. The conjecture's mechanism is discriminant-specific: disc -3 produces roots-of-unity arguments; disc -7 does not. The conjecture generalizes the disc -3 phenomenon to a discriminant -7 denominator, where it fails.

4 Exact verification

- `reproduce.py` (standard library only) recomputes: the roots and their invariants (Claim 1); S three independent ways agreeing to 6.4×10^{-14} ; the discriminant- -3 boundary sum; the exhaustive small-denominator search; and the exact fixed-point partial sums

$$\frac{1868497242085\dots 429833203580928}{1608225631910\dots 212099076096} = 1.161837745283165\dots \quad (N = 40),$$

$$\frac{1326774102455\dots 725602926375206912}{1133836264661\dots 7480348409856} = 1.170163756273512\dots \quad (N = 60),$$

the full 81- and 155-digit literals being reproduced verbatim by `reproduce.py` and asserted as theorems `psum40_val/psum60_val` in `lean4/Main.lean`.

- `lean4/Main.lean` (core Lean 4, toolchain v4.33.1, no Mathlib) formalizes the attack concretely in $\mathbb{Q}(\sqrt{-7})$ and $\mathbb{Q}(\sqrt{-3})$: r is a root of $z^2 + z + 2$; $|r|^2 = 2 \neq 1$; $r^3, \bar{r}^3, r^2, r \neq 1$; discriminants $-7 \neq -3$; the contrast $\omega^2 + \omega + 1 = 0$, $\omega^3 = 1$, $\omega\bar{\omega} = 1$; the partial-fraction mechanism sampled at $n = 0, 1, 2, 3$; and the exact fixed-point partial sums above. All 22 theorems are closed computations (`rfl/decide`); `Check.lean` audits `#print axioms` on each: *zero axioms, zero sorry*.